

Tarun Raheja

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EDUCATION

University of Pennsylvania

Master of Science in Scientific Computing; **GPA: 4.0/4.0**

Thesis Advisor : Prof Andrea Liu, **Topic** : Learning in Physical Systems

Philadelphia, USA

Aug. 2022 – Aug. 2024

Birla Institute of Technology and Science, Pilani

Bachelor of Engineering (Hons.) in Computer Science; **GPA: 8.57/10**

Thesis advisor : Prof. Sreejith GJ, **Topic** : DMRG for Potts' Model

Hyderabad, India

Aug. 2015 – Aug. 2019

Graduated with First Degree Honours

PUBLICATIONS

1. Lambert, Neill* and **Raheja, Tarun*** and Cross, Simon and Menczel, Paul and Ahmed, Shahnawaz and Pitchford, Alexander and Burgarth, Daniel and Nori, Franco : QuTiP-BoFiN: A bosonic and fermionic numerical hierarchical-equations-of-motion library with applications in light-harvesting, quantum control, and single-molecule electronics. *Phys. Rev. Res.* 5, 013181 (2023).
2. **Raheja, Tarun*** and Pochhi, Nilay* : One Guardrail Is Not Enough: A Security Game for Deploying Complementary LLM Defenses. In *17th Conference on Game Theory and AI for Security (GameSec 2026)*. (2026).
3. **Raheja, Tarun*** and Pochhi, Nilay* : One Crowd, Three Cascades: Information Diffusion in LLM-Agent Simulations Is Governed by Serialization, Not Society. In *COLM 2026 Second Workshop on Social Simulation with LLMs*. (2026).
4. **Raheja, Tarun*** and Pochhi, Nilay* : DimPFN: Units as an Exact Symmetry for In-Context Learning of Physical Laws. In *COLM 2026 Second Workshop on Language Models for Scientific Discovery*. (2026).
5. **Raheja, Tarun*** and Pochhi, Nilay* : One Posterior, Many Samplers: Open Problems in Sampling-Based Planning and Control. In *RSS 2026 Workshop on Sampling-Based Optimization in Robotics*. (2026).
6. **Raheja, Tarun*** and Pochhi, Nilay* : From RLHF to Direct Alignment: A Theoretical Unification of Preference Learning for Large Language Models. *arXiv:2601.06108* (2026).
7. **Raheja, Tarun*** and Pochhi, Nilay* : An initial exploration of using Persistent Homology for Noise-Resilient Tactile Object Recognition. In *NeurIPS 2024 Workshop on Touch Processing: From Data to Knowledge*. Oral Presentation (2024).
8. **Raheja, Tarun*** and Sinha, Raunak and Deepak, Advit and Healy, Will and Srinivasa, Jayanth and Lee, Myungjin and Kompella, Ramana : Enhancing Large Language Models through Transforming Reasoning Problems into Classification Tasks. In *Proceedings of the 2024 ACL Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pp. 6007–6016. ELRA and ICCL, Torino, Italia (2024).
9. Haputhanthri, Udith and Storan, Liam and Jiang, Yiqi and **Raheja, Tarun** and Shai, Adam and Akengin, Orhun and Miolane, Nina and Schnitzer, Mark J and Dinc, Fatih and Tanaka, Hidenori : Understanding and controlling the geometry of memory organization in RNNs. *arXiv:2502.07256 (Under review)*
10. **Raheja, Tarun*** and Pochhi, Nilay* : The Alignment Trilemma: A Theoretical Perspective on Recursive Misalignment and Human-AI Adaptation Dynamics. In *ICLR 2025 Workshop on Bidirectional Human-AI Alignment*. (2025).
11. **Raheja, Tarun*** and Pochhi, Nilay* : On the Limitations of Neural Networks for Option Pricing: Analysis of Volatility Regime Sensitivity. In *ICLR 2025: I Can't Believe It's Not Better: Challenges in Applied Deep Learning*. (2024).
12. **Raheja, Tarun*** and Pochhi, Nilay* : DAGnosis: A Benchmark for Complex DAG-Based Multi-Step Reasoning in Large Language Models. *(Under review)*
13. **Raheja, Tarun*** and Pochhi, Nilay* : Foundation Models Meet Continual Learning: Recent Advances, Challenges, and Future Directions. In *NeurIPS 2024 Workshop on Scalable Continual Learning for Lifelong Foundation Models*. Accepted (Poster) (2024).
14. **Raheja, Tarun*** and Pochhi, Nilay and Curie, F. : Recent Advancements in LLM Red-Teaming: Techniques, Defenses, and Ethical Considerations. *arXiv:2410.09097* (2024).

* Denotes first authorship

PATENTS

1. **Cache-Generated Frequently Asked Questions Page.**
T. Raheja, R. Sinha, W. Healy, J. Srinivasa, A. Deepak, R. R. V. R. Kompella, A. Payani.
US Patent 12,572,536 (Granted Mar. 2026).

2. **Automatic retrieval augmented generation with expanding context.**
A. Deepak, W. Healy, **T. Raheja**, J. Srinivasa, R. R. V. R. Kompella, R. Sinha.
US Patent App. 18/442,996 (2025).
3. **Memory-based function calling for logical formulation and inference for large language models.**
T. Raheja, A. Deepak, W. Healy, J. Srinivasa, G. Liu, R. R. V. R. Kompella, et al.
US Patent App. 18/439,966 (2025).
4. **Performance monitoring, mitigation, and retraining of large language models.**
A. Deepak, R. Sinha, W. Healy, **T. Raheja**, J. Srinivasa, R. R. V. R. Kompella.
US Patent App. 18/440,710 (2025).

RELEVANT EXPERIENCE

Accel

*AI Resident (to **Shekhar Kirani**)*

Bangalore, India

Feb. 2026 – Present

- Advising on **AI investments** and helping **portfolio companies**.

Kipo AI

Head of ML, Founding Machine Learning Engineer

San Francisco, USA

May. 2024 – Jan. 2026

- Built and owned ML and MLOps infra as the **first ML hire**, and Employee No. **2**.
- Architected and implemented a state-of-the-art scalable production **NLP** and **Vision** pipeline for understanding electronics datasheets. Outperforming competitors in **speed**, **cost**, and **data granularity**.
- Architected and implemented a state-of-the-art scalable production **agentic AI-search** pipeline for electronics components. This arbitrarily-fine-grained, domain-aware, **multi-modal**, **spec-accurate deep-research** is novel to the electronics industry.
- Architected and implemented tooling for **MLOps**, **DataOps**, and **training infra** for in-house vision and NLP models for several in-house tasks, specifically **YOLOv10** and **BERT** instances.
- Critically helped the company secure **2 major contracts**, directly worked on demos to clients, attended several sales **trade shows**. Mentored **2 interns**. Helped grow the team to 10 people, directly onboarded **2 engineers**.

Cisco Research

*Machine Learning Research Intern (with **Jayanth Srinivasa**)*

San Jose, USA

May. 2023 – Aug. 2023

- Led project on a **novel reasoning algorithm** for LLMs and an **LLM Gateway**. Resulted in **four patent submissions**.

Linguistic Data Consortium

*NLP Research Projects Programmer (with **Prof. Mark Liberman**)*

Philadelphia, USA

Sep. 2022 – Jan. 2024

- Worked on **forced alignment** and **generalized information representation schemes** for robust NLP models.

UPenn Department of Physics

*Graduate Research Assistant (with **Prof. Andrea Liu**, **Dr. Nachi Stern**)*

Philadelphia, USA

Jan. 2023 – May 2024

- Worked on algorithms for **learning in physical systems**. Uncovered novel **physical symmetry breaking** phenomena.

J.P. Morgan & Chase

*Quantitative Researcher (**QR Rates team**)*

Mumbai, India

Aug. 2021 – Aug. 2022

- Worked on developing mathematical pricing and volatility models for linear and nonlinear **rates derivatives** products.
- Worked on an OPA framework for **hedging residual Greeks**. Published docs for the **IR Future risk model**.
- Migrated internal **model usage restrictions**. Worked on two **high impact internal projects** reporting directly to MDs.

humit.app

Founding Backend Engineer

Bangalore, India

Feb. 2021 – Aug. 2021

- Built the social music sharing service, Humit. Designed and deployed several **core features** end-to-end including a **recommendation system**, and **async growth reporting services**.

RIKEN, Theoretical Quantum Physics Laboratory

*Research Assistant (with **Prof. Franco Nori**, **Dr. Neill Lambert**)*

Wako-shi, Japan

Oct. 2019 – Nov. 2020

- Developed an open source library to simulate **open quantum systems** using the **HEOM** formalism. Modelled the environment as a mixture of **fermionic** and **bosonic baths**. Studied **exceptional points** of large non-hermitian systems using a **novel block diagonalization** algorithm. Optimized **Noisy Mean Field Annealing** for NP hard problems using ideas from **quantum thermodynamics**.
- Worked on several research problems in **quantum information**, **open quantum systems**, and **combinatorial optimization**. Also worked on the **Coherent Ising Machine** project, and collaborated with **IBM Research Tokyo**.
- Worked on the admin team to **QuTiP**, an open-source package with **250k+ DAUs**.

tensoralpha GmbH

Quantitative Developer Intern (with Dr. Thomas Schmelzer)

- Worked on statistical analysis of proprietary Crypto derivatives trading strategies. Researched a novel **tracker strategy**.

Berlin, Germany

Jul. 2020 – Aug. 2020

Microsoft Research

Research Intern (with Dr. Karthik Ramachandra)

- Worked on **Project Froid**, on scalar **UDF inlining** for MSSQL. Improved if-else parsing from **exponential to linear**.

Bangalore, India

Jul. 2019 – Sep. 2019

HONORS & AWARDS

Exa AI Discovery Agents Hackathon 2025 : Won first place among 50+ teams.

Sentient Foundation Werewolf AGIthon (AGI House) 2024: Won first place in agent strategy/jailbreaking hackathon.

CS50 Puzzle Day 2020 (Harvard x Facebook) : Scored 8/8. Global winner.

WorldQuant Challenge : Achieved Gold ranking for high Sharpe, market-neutral, medium frequency alphas.

JEE Mains & Advanced : Ranked among top 0.3% nationwide, among 1.7 million candidates in the engineering entrance test.

Regional Mathematical Olympiad : Highest scorer in Bathinda region in India, for the first stage to the IMO.

Problem Solving Assessment Examination, CBSE : 99.8 percentile in national aptitude test, out of 1.5 million candidates.

ACADEMIC SERVICE, REVIEWING & MENTORING

Mentor, Conquest 2026 (BITS Pilani) : Asia's largest non-diluting startup accelerator

ICLR 2027

NeurIPS 2026

NeurIPS 2026 DynaFront Workshop

COLM 2026 LM4Sci Workshop

COLM 2026 Social Sim Workshop

ACM WebConf HCRS 2026

ICLR 2026

ICLR 2026 Workshops : I Can't Believe it's Not Better, MM Intelligence, P-AGI

ICLR 2025

NeurIPS 2025 Efficient Reasoning Workshop

NeurIPS 2025 Frontiers of Probabilistic Inference Workshop

NeurIPS 2024 RedTeaming GenAI Workshop

NeurIPS 2024 EvalEval Workshop

NeurIPS 2024 Workshop on Touch Processing

SKILLS

Programming : C, C++, Java, Python, MATLAB, R, SQL, $\text{\LaTeX}$, AWS, CI/CD

Technologies : Git, BASH, Django, Pandas, NumPy, PyTorch, TensorFlow, Spark, sklearn, Docker, Airflow

ML/SWE: Deep Learning, Computer Vision, Infrastructure, Data Structures, Algorithms, Scalable Design, Data Science